G11 PHYSICS

Assignment 02 Solutions

Problem 1. Test dimensionally if the formula $t = 2\pi\sqrt{\frac{m}{F/x}}$ may be correct, where t is the time period, m is mass, F is force and x is distance.

Solution. Since 2π is dimensionless, we only need to check the dimensions of $\sqrt{\frac{m}{F/x}} = \sqrt{\frac{mx}{F}}$.

We know,

$$[t] = T, \quad [m] = M, \quad [F] = MLT^{-2}, \text{ and } \quad [x] = L.$$

Then, the dimension of the right-hand side is

$$\left[\sqrt{\frac{mx}{F}}\right] = \sqrt{\frac{ML}{MLT^{-2}}} = \sqrt{T^2} = T.$$

The left-hand side is time and hence the dimension is T. The dimensions of both sides are equal and hence the formula *may* be correct.

Problem 2. Young's modulus Y of steel is $19 \times 10^{10}\,\mathrm{N/m^2}$. (Young's modulus measures the stiffness of a material and represents the ratio of stress to strain in the elastic region.) Express this value in $dyne/\mathrm{cm}^2$, where *dyne* is the CGS unit of force.

Solution. Looking at the units of Young's modulus Y, it is clear that its dimensional formula is

$$[\mathrm{Y}] = (MLT^{-2})(L^{-2}) = ML^{-1}T^{-2}.$$

Then,

$$1\,\mathrm{N/m^2} = (1\text{ kg})(1\text{ m})^{-1}(1\text{ s})^{-2},$$
$$1\mathrm{dyne/cm^2} = (1\text{ g})(1\text{ cm})^{-1}(1\text{ s})^{-2}.$$

Thus,

$$\frac{1\,\mathrm{N/m^2}}{1\,\mathrm{dyne/cm^2}} = \left(\frac{1\,\mathrm{kg}}{1\,\mathrm{g}}\right)\left(\frac{1\,\mathrm{m}}{1\,\mathrm{cm}}\right)^{-1}\left(\frac{1\,\mathrm{s}}{1\,\mathrm{s}}\right)^{-2} = \left(\frac{1000\,\mathrm{g}}{1\,\mathrm{g}}\right)\left(\frac{100\,\mathrm{cm}}{1\,\mathrm{cm}}\right)^{-1}\left(\frac{1\,\mathrm{s}}{1\,\mathrm{s}}\right)^{-2} = 10$$

or, $\quad 1\,\mathrm{N/m^2} = 10\,\mathrm{dyne/cm^2}$.

Hence,

$$19 \times 10^{10}\,\mathrm{N/m^2} = 19 \times 10^{11}\,\mathrm{dyne/cm^2}.$$

Problem 3. Suppose the distance covered by a particle in time t is given by

$$x = a + bt + ct^2 + dt^3.$$

Find the dimensions of a, b, c, and d.

Solution. Since x represents distance, it has the dimension $[x] = L$. Each term on the right-hand side must have the same dimension as x for the equation to be correct.

Therefore,

$$[a] = L$$

$$[bt] = L, \qquad \text{or,} \qquad [b] = LT^{-1}$$

$$[ct^2] = L, \qquad \text{or,} \qquad [c] = LT^{-2}$$

$$[dt^3] = L, \qquad \text{or,} \qquad [d] = LT^{-3}.$$

Problem 4. An important milestone in the evolution of the universe just after the Big Bang is the set of fundamental scales known as the *Planck units*. These include the Planck time t_p, the Planck length l_p, and the Planck mass m_p. Their values depend on three fundamental constants: (1) the speed of light (the fundamental constant of relativity), $c = 3.00 \times 10^8\,\mathrm{ms^{-1}}$; (2) Newton's gravitational constant (the fundamental constant of gravity), $G = 6.67 \times 10^{-11}\mathrm{m^3kg^{-1}s^{-2}}$; and (3) Planck's constant (the fundamental constant of quantum mechanics), $h = 6.63 \times 10^{-34}\mathrm{kg\ m^2s^{-1}}$.

(a) Based on dimensional analysis, find the value of the Planck time t_p. (This represents the smallest meaningful unit of time in physics, often interpreted as the time it takes light to travel one Planck length.)

(b) Repeat the derivation to obtain a quantity with the dimensions of length, and evaluate the result numerically. Ignore any dimensionless constants. (This is the Planck length, the smallest meaningful unit of length in physics.)

Solution.

(a) done in class

(b) Given:

$$c = 3.00 \times 10^8 \text{ m s}^{-1}, \qquad G = 6.67 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2}, \qquad h = 6.63 \times 10^{-34} \text{ kg m}^2\text{s}^{-1}.$$

Assume

$$l_p \propto G^a h^b c^c.$$

Since x represents distance, it has the dimension $[l_p] = L$. Then,

$$[L] = \left(M^{-1}L^3T^{-2}\right)^a \left(ML^2T^{-1}\right)^b \left(LT^{-1}\right)^c = M^{-a+b}\, L^{3a+2b+c}\, T^{-2a+-b-c}.$$

Equating powers of M, L, and T,

$$0 = -a + b \quad \Rightarrow \quad a = b, \tag{1}$$
$$1 = 3a + 2b + c, \tag{2}$$
$$0 = -2a - b - c. \tag{3}$$

Substituting eq. (1) in eq. (3), we get

$$0 = -2a - a - c \quad \Rightarrow \quad c = -3a. \tag{4}$$

Substituting eq. (1) and eq. (4) in eq. (2),

$$3a + 2a - 3a = 1 \quad \Rightarrow \quad a = \frac{1}{2}. \tag{5}$$

Then, substituting $a = \frac{1}{2}$ in eq. (1) and eq. (4),

$$b = \frac{1}{2}, \qquad c = -\frac{3}{2}.$$

Therefore,

$$\begin{aligned} l_p &\propto G^{1/2}h^{1/2}c^{-3/2} \\ &= \sqrt{\frac{Gh}{c^3}}. \end{aligned}$$

Ignoring dimensionless constants,

$$l_p = \sqrt{\frac{Gh}{c^3}} = \sqrt{\frac{(6.67 \times 10^{-11})(6.63 \times 10^{-34})}{(3.00 \times 10^8)^3}} \approx 4.05 \times 10^{-35}\text{ m}.$$

The actual value of Planck's length is $l_p \approx 1.616255 \times 10^{-35}$ m. Our result is a close approximation, differing by a factor of about 2.5. This highlights both the strength of dimensional analysis (it yields a value close to the true scale) and its limitation (it does not account for dimensionless constants).

Problem 5. To keep an object moving in a circle at constant speed requires a force called the centripetal force. This force always acts toward the center of the circular path and is responsible for continuously changing the direction of motion. Do a dimensional analysis of the centripetal force.

Solution.

Suppose the centripetal force F depends on the mass m, speed v, and radius r of the circular path. Then let

$$F \propto m^a v^b r^c.$$

Therefore,

$$[F] = [m]^a [v]^b [r]^c.$$

Using dimensions,

$$MLT^{-2} = M^a (LT^{-1})^b L^c = M^a L^{b+c} T^{-b}.$$

Equating powers of M, L, and T,

$$a = 1, \qquad b = 2, \qquad c = -1.$$

Hence,

$$F \propto \frac{mv^2}{r}.$$

Ignoring any dimensionless constant,

$$F = \frac{mv^2}{r}.$$

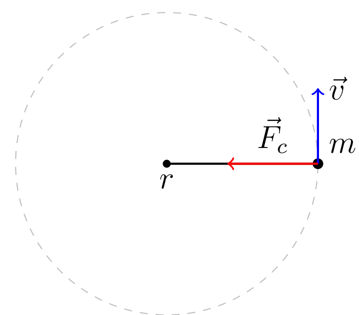

Fig.: Centripetal motion